Research & Expertise Navigator

Challenge analysis and team game plan — Hack Apertus, Liebefeld, 18 October 2026

In one sentence: build an expertise search engine that aggregates the public websites of UniBE, ARTORG and Diabetes Center Berne, understands natural-language questions (including wording that differs from the sites' own terminology) and returns **people, units, expertise, projects and infrastructure** — every result structured, explained and **linked to its source page**. Mandatory model: Apertus v1.5 (8B/70B). Sovereign deployment, outside the cantonal ICT.

1. What the Canton really wants (reading between the lines)

The brief is clear on one point: they do not want a chatbot that 'chats', they want **verifiable knowledge infrastructure**. The five evaluation criteria should be read as follows:

- **Purposeful use of AI** — AI only where it adds real value (structured extraction, terminology normalisation, match explanation), never as a decorative layer. Every place Apertus is used needs a stated reason.
- **Technical rigour** — reproducible pipeline, metrics measured on a test query set, demonstrably zero hallucinations. They will check: 80% recall@10, 100% traceability.
- **Value, cost & scalability** — cost per indexed page and per query; 8B on a single GPU as the baseline, 70B only where it improves quality. Show numbers.
- **Sovereign deployability** — everything in Docker, no foreign APIs, local open-source embeddings, data stays in Switzerland. This is a scoring criterion, not just a constraint: emphasise it in the pitch.
- **Implementation feasibility** — a credible path from MVP to production: adding a new site = one config block, no new code.

Key values to convey in the pitch: transparency (every claim has a URL), fairness (a small lab is as findable as a whole faculty), honesty (the system says 'not enough sources' instead of inventing), sovereignty (Swiss open weights, Apache 2.0, data never leaves CH), extensibility (config-driven).

2. Requirements, point by point (delivery checklist)

#	Brief requirement	What we must concretely show	How we measure it
R1	Finds the right people (≥80% of expected experts in top 10)	Golden set of 20–30 queries with expected experts (built from the pages themselves); hybrid ranking	Eval script: recall@10, printed in README and in the demo
R2	Handles synonyms and lay terms ('glucose sensors' → CGM, biosensing, wearable sensing)	Query expansion with Apertus + multilingual semantic embeddings (DE/EN/FR) + per-domain glossary	Synonym query pairs: same expert set in top 10
R3	100% traceable, no invented people/roles/projects	Every result field carries the source URL and the snippet it was extracted from; the LLM generates ONLY from retrieved chunks	Automatic validator: no result without source_url; 'non-existent person' test → negative answer
R4	Structured results (person, role/affiliation, unit, expertise, projects, infrastructure/services, match reason, link)	Fixed JSON schema; fields left empty if the source does not provide them (never filled by inference)	Schema validation (pydantic) on every result
R5	Covers all three sources: UniBE + ARTORG + DCB	Crawl configured for the 3 domains; source filter in the UI	Page/entity count per source on the dashboard
R6	Easy to extend	New source = new YAML block (domains, seed URLs, optional selectors, language); README with a 3-step procedure	Live demo: add a 4th site in <5 minutes

#	Brief requirement	What we must concretely show	How we measure it
R7	Says when sources are insufficient	Confidence threshold on retrieval; below threshold → explicit 'no verifiable source' message with suggestions	Out-of-domain queries in the test set → abstention
R8	Sovereign / air-gapped	Single docker compose: crawler, vector DB, Apertus via vLLM, API, UI. No external calls after the crawl	Start with network disabled: it works
R9	nDSG/GDPR, public content only, Open Commons	Respect robots.txt, public pages only, no data beyond what is already published, open licence on the repo	'Compliance' section in the README

3. Our proposed solution

Architecture: a 5-stage pipeline, fully in Docker, with Apertus used at 3 precise points.

Stage	What it does	Tooling	Role of Apertus
1. Discovery & crawl	Starts from the unibe.ch subdomain inventory + ARTORG and DCB seeds; follows in-domain links; respects robots.txt; stores raw HTML with URL and timestamp	httpx/Scrapy; trafilatura for clean text	—
2. Page classification	Detects page type: person / group-institute / project / infrastructure-service / other	URL + template rules (fast) + Apertus 8B fallback	Classifies ambiguous pages
3. Structured extraction	Extracts entities per the schema: name, role, affiliation, public email, expertise, projects, infrastructure; every field with its source snippet	Apertus 8B with constrained JSON output (JSON schema via vLLM)	**Main use.** Extraction with mandatory snippet citation
4. Enrichment & indexing	Normalises expertise terms (synonyms, lay terms, DE/EN/FR), computes embeddings; builds person↔unit↔project↔infrastructure graph	Local open multilingual embeddings (e.g. bge-m3 / multilingual-e5); Qdrant; BM25 for keywords	Generates terminology variants and glossary per expertise
5. Search & answer	NL or keyword query → synonym expansion → hybrid search (BM25 + vectors) → rerank → group by person/unit → explain match citing snippets → abstain if score is low	FastAPI + lightweight UI; results as structured cards + exportable JSON	Expands the query; writes 'why it matches' ONLY from retrieved chunks

Why this architecture wins

- **Strict separation of extraction and search:** the LLM never 'answers from memory'; every output originates from a chunk with a URL. A direct answer to the traceability criterion and the ethical guidelines.
- **Hybrid search:** BM25 covers the sites' exact terms, embeddings cover synonyms and lay language, Apertus expansion covers domain terms (CGM, biosensing). Together they meet criterion 2 without fine-tuning.
- **Lightweight knowledge graph:** person→group→institute→faculty lets us return 'people and units' and gives fair visibility to small groups (ranking by relevance, not by site size).
- **Config-driven:** *sources.yaml* is the only thing to touch to add an institution. Demonstrable live.
- **Sovereign and low-cost:** Apertus 8B in FP8 runs on a single ~16 GB GPU; indexing is a one-off batch, search is nearly free. 70B is optional for refining extraction if GPUs are available.

Result schema (fixed contract)

```
{ person, role, affiliation: {group, institute, faculty}, expertise: [..], projects: [{title, url}], infrastructure_services: [{name, url}], why_match: 'short cited text', sources: [{url, snippet, retrieved_at}], confidence }
```

Golden rule: a field not supported by a snippet stays empty. Never 'complete' it with the model's own knowledge.

4. Work plan and role split

Goal: a working end-to-end demo as early as possible, then improve quality and metrics. Never reach the pitch without numbers.

Phase	Activities	Suggested owner	Deliverable
0 — Setup (first 2 h)	Repo, docker compose, vLLM serving Apertus 8B (image ghcr.io/swiss-ai/vllm_apertus_1.5_release), vector DB, sources.yaml with the 3 domains	Infra dev	Stack starts with one command
1 — Crawl (h 2–6)	Crawler on the 3 domains with limits (depth, max pages); priority to 'team', 'people', 'research', 'projects', 'facilities' pages	Data dev	Corpus of clean pages with URLs
2 — Extraction (h 4–10)	Prompt + JSON schema for Apertus; hand-check on 30 pages; pydantic validator; mandatory snippets	AI dev	Entity table with sources
3 — Index & search (h 8–14)	Embeddings, BM25, hybrid search, rerank, grouping by person; query expansion; abstention threshold	AI dev + backend dev	Working /search API
4 — Golden set & eval (h 10–16)	20–30 queries with expected experts (from real pages), incl. synonym/lay-term pairs and 3 out-of-domain queries; eval script	Everyone (10 min each)	recall@10, 100% traceability, correct abstention
5 — UI (h 12–18)	Search bar, result cards with fields and links, source filter, 'verified source' badge, 'export JSON' button, 'insufficient sources' state	Frontend dev	Clickable demo
6 — Extensibility (h 16–19)	Add a 4th site via YAML, recorded on video / live; README with the 3 steps	Infra dev	Living proof of criterion 6
7 — Pitch (last 3 h)	Slides, scripted demo on 3 queries, eval numbers, costs, roadmap	Mattia + 1	5-min pitch + demo

5. Risks and mitigations

Risk	Mitigation
Crawl too broad (unibe.ch has dozens of subdomains) and we run out of time	Cap at ~2,000 pages total; person/group pages first; ARTORG and DCB crawled fully (they are small) to get 100% coverage on two of three sources
Apertus 8B produces invalid JSON or invents fields	Schema-constrained output; retry at temperature 0; validator drops fields without a snippet; log of dropped fields shown as proof of rigour
Pages in German/French, queries in English	Multilingual embeddings + query expansion in DE/EN/FR; the brief rewards this as 'handles different wording'
GPU unavailable or slow	Quantised Apertus 8B (FP8/NVFP4) or Q4 via llama.cpp for batch extraction; search does not depend on real-time LLM if 'why match' candidates are pre-computed
Live demo fails	3 queries rehearsed 20 times, index pre-built, fallback to screenshots/recording

6. Pitch structure (5 minutes)

- **0:00 The problem in 20 seconds** — 'glucose sensors' vs 'CGM': today you must know where to search before you know what to search for.

- **0:30 Live demo, 3 queries** — (1) lay term, (2) same need with technical term → same experts, (3) query with no sources → the system honestly abstains. Every card shows link and snippet.
- **2:30 How it works** — one picture of the 5-stage pipeline; where and why we use Apertus; all in Docker, no data leaves Switzerland.
- **3:30 The numbers** — recall@10 on the golden set, 100% results with a source, pages indexed per source, estimated cost per 10,000 pages and per query on a single GPU.
- **4:15 Extensibility and roadmap** — 'adding Inselspital = 12 lines of YAML'; next steps: incremental refresh, API for cantonal portals, researchers' feedback on their own profiles.
- **4:45 Close on values** — transparent, fair, honest, sovereign, open (Open Commons).

Do NOT: promise 'natural-language answers' without sources; use external APIs (OpenAI, Google) even just for embeddings; present results without the evaluation set; show personal data not already public on the sites.

Quick references

Model: swiss-ai/Apertus-v1.5-8B and -70B (Apache 2.0, 262K-token context, multilingual, tool calling and JSON output supported via vLLM with the image ghcr.io/swiss-ai/vllm_apertus_1.5_release). Data sources: supplied unibe.ch subdomain inventory, artorg.unibe.ch, Diabetes Center Berne website. Constraints: deployment outside the Canton of Bern's ICT, nDSG/GDPR, Open Commons outputs (T&C §6).